

Technology Stack

Date	29 September 2026
Team ID	UNM1659GenieBirds
Project Name	EduGenie – AI-Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

EduGenie is an AI-powered learning assistant that provides Ask a Question, Explain a Topic, Generate Quiz, Summary, and Recommended Learning Path features. The selected stack supports a simple interactive interface, Python-based application logic, AI integration, and future scalability.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Provides a simple interactive web interface for learners. It supports forms, buttons, text input, and readable AI-generated results with minimal frontend code.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
2	Backend / Server-Side	Python	Handles application logic, feature workflows, prompt preparation, response processing, validation, and integration with the AI service.
3	Database / Data Storage	SQLite	Provides lightweight relational storage for learner information, interaction history, quiz data, and other structured project data while keeping the prototype simple.
4	Cloud / Hosting / Deployment	Streamlit Community Cloud	Provides straightforward web deployment for the Streamlit application and supports easy sharing of the working prototype for demonstration and testing.
5	Version Control & CI/CD	Git + GitHub	Supports collaborative development, version history, code backup, and controlled updates. GitHub also provides a central repository for the EduGenie source code.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools	Google Gemini API	Provides the generative AI capability used to create answers, explanations, summaries, quizzes, and personalized learning recommendations from learner prompts.

Technology Stack Summary

The stack is intentionally lightweight and suitable for the current EduGenie prototype. It separates the user interface, application logic, data storage, deployment, version control, and AI integration so the solution can be extended later with stronger authentication, scalable database services, caching, monitoring, and additional learning resources.